

## PROJECT #5 – INTEGRATE

Project #5 will look slightly different for each student group as you will each be working on a different project. However, irrespective of your project topic, all student groups will be asked to complete the following tasks. For more information about your specific project client and requirements, please see the information below.

Regardless of project topic, groups need to complete the following tasks:

- Coding Skills:
  - Install and use at least three packages of their choice (can be from GitHub)
  - Plot at least 2 graphs using the appropriate graph for the data type
  - Define and use a function within their code
  - Clean and organize data from an external source
  - Discuss coding choices made through the analysis to show they understand there is bias associated with how decisions are made
- Complete all of the relevant 'engineering' skill building activities from the semester
  - Scope of work
  - Gantt Chart
  - Timesheet
  - Technical and non-technical executive summaries
  - Analysis of bias/ethics in coding (covered above)

**Project #5 will be due Friday May 8<sup>th</sup> at 5:00pm.**

### Project Topic:

## TRANSPORTATION

### *Project #1: Autonomous Vehicle Interactions at an Intersection (Autonomous Vehicles)*

As the U.S. advances toward a future transportation system that includes autonomous vehicles (AVs) operating alongside traditional road users, understanding how these agents interact is critical. The U.S. Department of Transportation (USDOT) has tasked your team with analyzing a subset of the **Third Generation Simulation (TGSIM)** dataset -- specifically, the **Foggy Bottom** segment -- to study agent behavior around intersections.

Your objective is to evaluate the claim that AVs are generally safer and more predictable than other agents by analyzing 200 seconds of recorded trajectory data.

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| <b>Client</b>          | Nebraska Department of Transportation (NDOT)                                                                                                                                                                                                                                                                                                |
| <b>Client requests</b> | 1. A <b>quantitative comparison</b> of the behavior of different transportation agents (e.g., AVs, non-automated passenger vehicles, pedestrians). Given the limited 200-second duration of the dataset, comparing complete trajectories may not be meaningful. Instead, focus on localized interactions or behavior patterns. For example: |

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|                     | <ul style="list-style-type: none"><li>• Compare acceleration and deceleration patterns within intersection zones or near pedestrian areas approached by Avs.</li><li>• Analyze speed profiles along specific lanes to detect consistency or variability between agents.</li><li>• Measure and compare average speeds inside intersections or count the number of deceleration events, which may indicate disruptive behavior.</li><li>• These are just a few possibilities, and there are many behavioral insights that can be extracted from this rich dataset.</li></ul> <p>Present your findings using clear summary tables and visualizations that support your interpretations.</p> <p>2. A <b>visual representation</b> of agent movement at the intersection, either as a static plot of location vs. time or an animation that highlights how the agents interact over time.</p> |
| Data sources        | <ul style="list-style-type: none"><li>• <a href="#">Transformed TGSIM Foggy Bottom 200sec.csv</a></li><li>• <a href="#">TGSIM Data Dictionary.xlsx</a></li><li>• Sample Python code for plotting intersection polygons: <a href="#">Polygons.ipynb</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Report requirements | <ol style="list-style-type: none"><li>1. <b>Dataset Description</b><br/>Describe the dataset in detail, for example: the types of information available, the data structure and organization, key data elements, any preprocessing applied.</li><li>2. <b>Analysis and Insights</b><ol style="list-style-type: none"><li>a. Present your behavioral analysis of the different agents.</li><li>b. Comment specifically on the behavior of AVs relative to other agents.</li><li>c. Support your discussion with appropriate figures, plots, or animations, and provide a thoughtful interpretation of the findings.</li></ol></li></ol>                                                                                                                                                                                                                                                   |